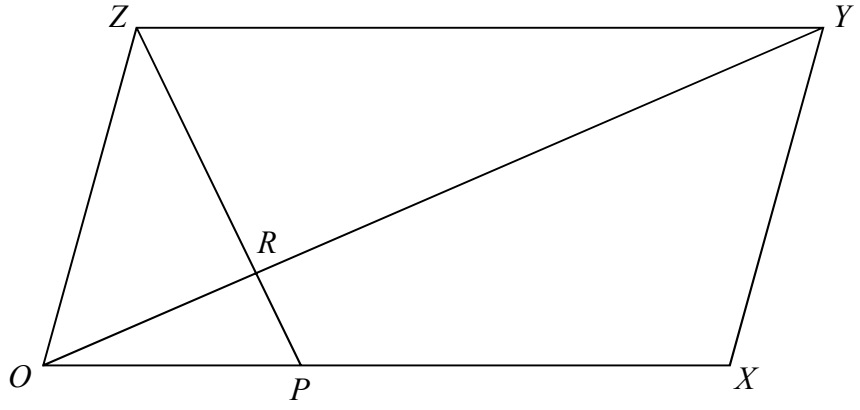
24 $OXYZ$ is a parallelogram.



$\overrightarrow{OX} = \mathbf{a}$

$\overrightarrow{OY} = \mathbf{b}$

P is the point on OX such that $OP : PX = 1 : 2$
R is the point on OY such that $OR : RY = 1 : 3$

Work out, in its simplest form, the ratio $ZP : ZR$
You must show all your working.

OP = (1/3)a and OR = (1/4)b.

Since OXYZ is a parallelogram, OZ = b–a.

ZP = OP–OZ = (1/3)a–(b–a)

= (4/3)a–b.

ZR = OR–OZ = (1/4)b–(b–a)

= a–(3/4)b.

Thus ZP = (4/3)ZR, so ZP:ZR = 4:3.

5 marks total
Fractional vectors
Find OZ
Form ZP, ZR
Common vector multiple
Ratio 4:3

(Total for Question 24 is 5 marks)